

# Global optimization of common acquisition functions is NP-hard

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## Abstract

We study the computational complexity of minimizing the posterior mean of a Gaussian process (GP) with zero prior mean, and of several standard Bayesian optimization acquisition functions built from that same GP. We show that for every stationary, isotropic, strictly positive definite kernel in a broad family that includes the squared-exponential (SE) kernel and *every* half-integer Matérn kernel  $\nu = \frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \dots$ , minimizing the posterior mean over  $[-1, 1]^d$  is NP-hard, via a polynomial-time reduction from MAX-CUT. The reduction embeds the MAX-CUT objective exactly at the vertices of the hypercube, then repairs a weakness of exact vertex agreement to control the interior of the box with a diagonal concavity shift and a polynomial-size kernel-expansion approximation, so that the extreme-point theorem for concave functions on a polytope pins the global minimum to a vertex. We then show the hardness is not an artifact of the posterior mean alone: it persists for the lower confidence bound, expected improvement, and the uniform-cost Gittins index acquisition functions, via a single mechanism common to all of them. A closing remark discusses how much further the argument generalizes, and where it provably does not.

## Contents

|          |                                                                                 |           |
|----------|---------------------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Background and existing literature</b>                                       | <b>2</b>  |
| <b>2</b> | <b>Preliminaries</b>                                                            | <b>2</b>  |
| 2.1      | Gaussian process regression . . . . .                                           | 2         |
| 2.2      | The kernel family . . . . .                                                     | 3         |
| <b>3</b> | <b>NP-hardness of minimizing the posterior mean over <math>[-1, 1]^d</math></b> | <b>3</b>  |
| 3.1      | MAX-CUT . . . . .                                                               | 4         |
| 3.2      | A concave reformulation . . . . .                                               | 4         |
| 3.3      | Far-field tools: a kernel-independent distance expansion . . . . .              | 4         |
| 3.4      | Directional bump lemmas . . . . .                                               | 5         |
| 3.5      | Polynomial-size approximation, unified . . . . .                                | 8         |
| 3.6      | Rational realizability, unified . . . . .                                       | 8         |
| 3.7      | Main theorem . . . . .                                                          | 10        |
| <b>4</b> | <b>Scope: how general is this?</b>                                              | <b>11</b> |
| <b>5</b> | <b>Extension to acquisition functions</b>                                       | <b>12</b> |
| 5.1      | A domination lemma . . . . .                                                    | 12        |
| 5.2      | Lower confidence bound . . . . .                                                | 12        |
| 5.3      | Expected improvement . . . . .                                                  | 13        |
| 5.4      | Gittins index . . . . .                                                         | 14        |

## 1 Background and existing literature

Bayesian optimization has long been a useful way to optimize functions which are costly to evaluate. Using a Gaussian process prior, even a few observations can give enough information to change one's beliefs greatly. The primary question for the practitioner is to decide where to evaluate the function next after making some observations. A number of acquisition functions exist to represent how "promising" each potential point is as a possible improvement upon the data the user already has. In recent years, some commonly used ones have been the following:

- Expected improvement: The expected positive part of the improvement over the current running minimum at each point. If we can only make a single observation, this is optimal; otherwise, we can view it as a greedy approximate dynamic programming algorithm.
- Lower confidence bound: The infimum of a posterior credible interval of the function value at each point. This comes from the multi-armed bandits literature ([3]).
- Gittins index: Tailored to cases where there is a cost to make each observation rather than a fixed observation budget, the Gittins index represents gains relative to cost and includes the option for the user to decline to make an observation if no points are worth the cost. Introduced by [4], followed up on in [5], and introduced to Bayesian optimization in [10].

Surprisingly little attention has been given to the question of how one would actually optimize these functions. [9] proposes a greedy approach for these and a near-optimal Monte Carlo estimation scheme for a different class of acquisition functions. More recently, [8] offers a promising surrogate-based branch-and-bound approach, and [11] proposes a mixed-integer quadratic programming formulation, both of which would take exponential time in the worst case. The question of whether a polynomial-time algorithm could succeed globally has not been addressed formally in the literature. We prove here that for the most commonly used covariance kernels, the problem is NP-hard, meaning that there can be no polynomial-time algorithm to solve the problem unless  $P=NP$ . We do this via a reduction from MAX-CUT, already known to be NP-hard (see [2]). The proof is therefore similar in structure to the proof in [6] that quadratic programming is NP-hard when the quadratic form is indefinite.

## 2 Preliminaries

### 2.1 Gaussian process regression

Let  $f \sim \text{GP}(0, k)$  be a Gaussian process on  $\mathbb{R}^n$  with zero prior mean and covariance kernel  $k : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ . Given training inputs  $X = \{p_1, \dots, p_m\} \subset \mathbb{R}^n$ , targets  $y \in \mathbb{R}^m$ , and observation noise variance  $\sigma_n^2 \geq 0$ , write  $K \in \mathbb{R}^{m \times m}$  for the Gram matrix  $K_{lj} = k(p_l, p_j)$  and  $k(x, X) \in \mathbb{R}^m$  for the vector with entries  $k(x, p_l)$ . The posterior mean and variance of  $f$  at a query point  $x$  are

$$\mu(x) = k(x, X)^\top (K + \sigma_n^2 I)^{-1} y = \sum_{l=1}^m \alpha_l k(x, p_l), \quad \alpha := (K + \sigma_n^2 I)^{-1} y, \quad (1)$$

$$\sigma^2(x) = k(x, x) - k(x, X)^\top (K + \sigma_n^2 I)^{-1} k(x, X). \quad (2)$$

Two facts about (1)–(2) are used repeatedly and without further comment: (a) for *any* target dual coefficients  $\alpha \in \mathbb{R}^m$  and any  $\sigma_n^2 \geq 0$ , setting  $y := (K + \sigma_n^2 I)\alpha$  realizes  $\mu(x) = k(x, X)^\top \alpha$  exactly, so any desired dual vector is realizable as a genuine posterior mean; (b)  $\sigma^2(x)$  does not depend on  $y$  (or on  $\alpha$ ) at all — only on  $X$ ,  $k$ , and  $\sigma_n^2$  — and satisfies  $0 \leq \sigma(x) \leq \sigma_f$  whenever  $k(x, x) = \sigma_f^2$  for all  $x$ , since (2) is a difference of a fixed diagonal value and a nonnegative quadratic form (as  $K + \sigma_n^2 I \succ 0$  is a genuine covariance-of-observations matrix).

## 2.2 The kernel family

**Definition 2.1** (Stationary isotropic kernel). *A kernel  $k$  is stationary and isotropic (radial) with length scale  $\ell > 0$  if  $k(x, x') = \sigma_f^2 \rho(\|x - x'\|/\ell)$  for a fixed constant  $\sigma_f^2 > 0$  and a fixed shape function  $\rho : [0, \infty) \rightarrow (0, 1]$  with  $\rho(0) = 1$ . It is strictly positive definite (PD) if for every finite set of pairwise distinct points  $p_1, \dots, p_m \in \mathbb{R}^n$  (any  $n$ ), the Gram matrix  $k(p_i, p_j)$  is positive definite.*

Throughout we take  $\sigma_f = 1$  without loss of generality. Two named families instantiate Definition 2.1.

- Squared exponential (SE):  $\rho_{\text{SE}}(s) = \exp(-s^2/2)$ .
- Half-integer Matérn- $\nu$ ,  $\nu = p + \frac{1}{2}$ ,  $p \in \mathbb{N} \cup \{0\}$ :

$$\rho_\nu(\zeta) = Q_p(\zeta)e^{-\zeta}, \quad Q_p(\zeta) := \frac{p!}{(2p)!} \sum_{i=0}^p \frac{(p+i)!}{i!(p-i)!} (2\zeta)^{p-i}, \quad \zeta := \frac{\sqrt{2\nu} \|x - x'\|}{\ell}.$$

Both families are strictly PD in every dimension  $d$  ([7]).

Section 3 shows minimizing the posterior mean over  $[-1, -1]^d$  is NP-hard for the SE kernel and for *every* half-integer Matérn kernel at once. Section 5 extends this to LCB, EI, and the Gittins index. Section 4 discusses how much further the mechanism reaches.

## 3 NP-hardness of minimizing the posterior mean over $[-1, 1]^d$

Our proof is based on reducing the posterior mean minimization problem to a known NP-hard problem from graph theory. As such, we will need to express the problem in terms of a graph. We define the following terms:

- $G = (V, E)$  is a graph with  $|V| = d$  and edge weights  $w : E \rightarrow \mathbb{N}$ .
- The total weight of the graph is  $W_{\text{tot}} := \sum_{(i,j) \in E} w_{ij}$ .
- For  $(i, j) \in E$ , we define the weight matrix  $W$  such that  $W_{ij} = W_{ji} = \frac{1}{2}w_{ij}$ .
- The quadratic form of  $W$  is  $Q(u) := u^\top W u$  for  $u \in \{-1, 1\}^d$ , and its minimum is  $Q^*$ .
- The cut value of  $u$  is  $C(u) = \frac{1}{2}(W_{\text{tot}} - Q(u))$ .
- $\Lambda := \frac{1}{2}W_{\text{tot}}$  is an upper bound on  $\lambda_{\max}(W)$  by Gershgorin's theorem.
- $\tilde{Q}(x) := x^\top (W - \Lambda I)x$  for  $x \in \mathbb{R}^d$  is a shift of  $Q$ , and it is concave because  $\Lambda \geq \lambda_{\max}(W)$ .

### 3.1 MAX-CUT

MAX-CUT is the problem of computing  $C^* = \max_u C(u)$ . Determining whether  $C^* \geq K$  is NP-complete, as is unweighted MAX-CUT, the version with  $w_{ij} \equiv 1$  ([2], problem ND16). We will use this fact in the box-extension construction of §3.2–§3.5 since  $W_{\text{tot}} = \text{poly}(d)$ . Weighted instances with  $W_{\text{tot}} = \text{poly}(d)$  are explicitly mentioned too (Remark 3.13).

### 3.2 A concave reformulation

Our strategy will be to introduce  $\tilde{Q}$  as a diagonal shift to  $Q$  before ever introducing the kernel.

**Lemma 3.1** (Concave shift). *We have  $\min_{[-1,1]^d} \tilde{Q} = \min_{\{-1,1\}^d} \tilde{Q} = Q^* - \Lambda d$ .*

*Proof.* At a vertex,  $\|x\|^2 = d$ , so  $\tilde{Q}(x) = x^\top W x - \Lambda \|x\|^2 = Q(x) - \Lambda d$ . Since  $[-1,1]^d$  is a compact convex polytope, a concave function on it attains its minimum at an extreme point, yielding our desired result.  $\square$

This reduces the box problem to approximating a single, fixed, explicit concave quadratic  $\tilde{Q}$  well enough on *all* of  $[-1,1]^d$ , not just at the vertices. **Standing assumption for the rest of this section:**  $G$  is an unweighted MAX-CUT instance ( $w_{ij} \equiv 1$ ). Remark 3.13 below records the extension to weighted instances with  $W_{\text{tot}} = \text{poly}(d)$ .

Expanding

$$\tilde{Q}(x) = \sum_{(i,j) \in E} x_i x_j - \Lambda \sum_i x_i^2,$$

since

$$x_i x_j = \frac{1}{2} (v_{ij}^{+\top} x)^2 - \frac{1}{2} (v_{ij}^{-\top} x)^2$$

for unit vectors  $v_{ij}^\pm := (e_i \pm e_j)/\sqrt{2}$ , we have

$$\tilde{Q}(x) = \sum_{k=1}^{d+2|E|} \gamma_k (v_k^\top x)^2,$$

a sum of  $O(d^2)$  squared linear functionals of  $x$  along unit directions  $v_k \in \{e_i\} \cup \{v_{ij}^\pm\}$ , with coefficients  $\gamma_k \in \{-\Lambda\} \cup \{\pm \frac{1}{2}\}$ . Since

$$\Lambda = \frac{1}{2}|E| \leq \frac{1}{2} \binom{d}{2} \leq \frac{d^2}{4},$$

we have

$$|\gamma_k| \leq \Gamma := \max \left\{ \Lambda, \frac{1}{2} \right\} = \frac{|E|}{2} \leq \frac{d^2}{4} \quad \text{for all } k.$$

### 3.3 Far-field tools: a kernel-independent distance expansion

The two-point “directional bump” construction of the next subsection places training points far outside  $[-1,1]^d$ . The following elementary geometric fact controls the distance from a far-away point to anywhere in the box, and depends on no kernel at all. We will use this in every Matérn family bump lemma.

**Lemma 3.2** (Far-field distance expansion). *Let  $z \in \mathbb{R}^d$  with  $\|z\| \leq D$ , let  $v \in \mathbb{R}^d$  be a unit vector,  $t := v^\top z$ , and  $R \geq 2D$ . Then  $r_\pm := \|z \mp Rv\|$  satisfies  $r_\pm = R \mp t + \eta_\pm$  with  $|\eta_\pm| < 2D^2/R$ .*

*Proof.* We have

$$r_{\pm}^2 = R^2 \mp 2Rt + \|z\|^2 = R^2(1 + u_{\pm}) \quad \text{with} \quad u_{\pm} := \mp \frac{2t}{R} + \frac{\|z\|^2}{R^2}.$$

Since  $R \geq 2D \geq 2|t|$ , we have  $u_{\pm} \in [-3/4, 5/4]$ , and on this interval,

$$\left| \sqrt{1+u} - 1 - \frac{u}{2} \right| \leq \frac{2u^2}{9},$$

so

$$\left| r_{\pm} - R - \frac{Ru_{\pm}}{2} \right| \leq \frac{2Ru_{\pm}^2}{9}.$$

Since

$$\frac{Ru_{\pm}}{2} = \mp t + \frac{\|z\|^2}{2R} \quad \text{and} \quad \frac{2Ru_{\pm}^2}{9} \leq \frac{2R}{9} \left( \frac{2D}{R} + \frac{D^2}{R^2} \right)^2 \leq \frac{25D^2}{18R} \quad \text{for} \quad R \geq 2D,$$

collecting the two error sources gives

$$|\eta_{\pm}| \leq \frac{D^2}{2R} + \frac{25D^2}{18R} = \frac{17D^2}{9R} < \frac{2D^2}{R}.$$

□

### 3.4 Directional bump lemmas

We give two directional-bump lemmas: an exact one for SE (which needs no far-field expansion at all, since SE depends on the *squared* distance, whose exact expansion has no remainder), and a single lemma covering the half-integer Matérn kernels.

**Lemma 3.3** (Directional bump, SE). *Let  $k = k_{\text{SE}}$ . For a unit vector  $v \in \mathbb{R}^d$ ,  $\gamma \in \mathbb{R}$ , and  $\ell \geq 4d$ , set*

$$\psi_{v,\gamma}(x) := \gamma \ell e^{\ell/2} [k(x, \ell^{3/2}v) + k(x, -\ell^{3/2}v)].$$

*Then for every  $x \in [-1, 1]^d$ ,*

$$|\psi_{v,\gamma}(x) - 2\gamma\ell - \gamma(v^{\top}x)^2| \leq \frac{5|\gamma|d^2}{\ell}.$$

*Proof.* Setting  $t := v^{\top}x/\sqrt{\ell}$  and expanding quadratic terms, we find

$$k(x, \ell^{3/2}v) + k(x, -\ell^{3/2}v) = 2 \exp\left(-\frac{\|x\|^2 + \ell^3}{2\ell^2}\right) \cosh(t),$$

so

$$\psi_{v,\gamma}(x) = 2\gamma\ell \exp\left(-\frac{\|x\|^2}{2\ell^2}\right) \cosh(t).$$

Using the Taylor expansions

$$e^{-s} \leq 1 - s + \frac{s^2}{2} \text{ for } s \geq 0, \quad \cosh(t) \leq 1 + \frac{t^2}{2} + t^4 \text{ for } |t| \leq \frac{1}{2},$$

the product satisfies

$$\exp\left(-\frac{\|x\|^2}{2\ell^2}\right) \cosh(t) = 1 + \frac{t^2}{2} - \frac{\|x\|^2}{2\ell^2} + \text{err}, \quad |\text{err}| \leq \frac{2d^2}{\ell^2},$$

since each of the five cross-terms is  $O(d^2/\ell^2)$  or smaller. This gives

$$\psi_{v,\gamma}(x) = 2\gamma\ell + \gamma(v^\top x)^2 - \frac{\gamma}{\ell}\|x\|^2 + 2\gamma\ell \cdot \text{err},$$

and

$$\left| -\frac{\gamma}{\ell}\|x\|^2 + 2\gamma\ell \cdot \text{err} \right| \leq \frac{|\gamma|d}{\ell} + 4|\gamma|\ell \cdot \frac{d^2}{\ell^2} \leq \frac{5|\gamma|d^2}{\ell}.$$

□

**Lemma 3.4** (Degree stability under differentiation). *For a given half-integer  $\nu = p + \frac{1}{2}$ , let  $\rho_\nu^{(j)}$  be the  $j$ th derivative of  $\rho_\nu$  and define  $Q^{(j)}(\zeta) = \rho_\nu^{(j)}(\zeta)e^\zeta$ . Every  $Q^{(j)}$  has degree  $p$  and leading coefficient  $\frac{(-1)^j p! 2^p}{(2p)!} = \frac{(-1)^j}{(2p-1)!!}$ .*

*Proof.* We proceed by induction on  $j$ . As the base case, we note that  $Q^{(0)} = Q_p$ , which has leading term  $\frac{p! 2^p}{(2p)!} \zeta^p$ . Assuming the hypothesis for a given  $j$ , differentiating  $\rho_\nu^{(j)} = Q^{(j)}e^{-\zeta}$  and using the definition of  $Q^{(j+1)}$  gives

$$\rho_\nu^{(j+1)} = (Q^{(j)})' - Q^{(j)}e^{-\zeta} = Q^{(j+1)}e^{-\zeta} \Rightarrow Q^{(j+1)} = (Q^{(j)})' - Q^{(j)}.$$

Since  $Q^{(j+1)'}$  either has degree  $p-1$  or is identically zero, the leading term is the leading term of  $Q^{(j)}$  times  $-1$ , implying our desired result. □

**Lemma 3.5** (Polynomial envelope). *There exist finite constants  $A(p) \geq 1$ ,  $a(p) > 0$ , and  $\zeta_1(p) \geq 2$ , depending only on  $p$ , such that for all  $\zeta \geq 0$  and  $0 \leq j \leq 4$ ,  $|Q^{(j)}(\zeta)| \leq A(p)(1+\zeta)^p$ , and for all  $\zeta \geq \zeta_1(p)$ ,  $|Q^{(2)}(\zeta)| \geq a(p)\zeta^p$ .*

*Proof.* Each  $Q^{(j)}$  has degree  $p$  by Lemma 3.4, so  $Q^{(j)}(\zeta)/(1+\zeta)^p$  is continuous on  $[0, \infty)$  and tends to a finite limit as  $\zeta \rightarrow \infty$ , so it is bounded. Take

$$A(p) := 1 + \max_{0 \leq j \leq 4} \sup_{\zeta \geq 0} \frac{|Q_j(\zeta)|}{(1+\zeta)^p}.$$

Since  $Q^{(2)}(\zeta)/\zeta^p \rightarrow \frac{1}{(2p-1)!!} > 0$  as  $\zeta \rightarrow \infty$ , there exists  $\zeta_1(p) \geq 2$  such that

$$\left| \frac{Q^{(2)}(\zeta)}{\zeta^p} - \frac{1}{(2p-1)!!} \right| < \frac{1}{2(2p-1)!!} \quad \forall \zeta \geq \zeta_1(p),$$

so  $|Q^{(2)}(\zeta)| > \frac{1}{2(2p-1)!!} \zeta^p$  in that case. Take  $a(p) := \frac{1}{2(2p-1)!!}$ . □

**Lemma 3.6** (Directional bump, every half-integer Matérn kernel). *Let  $\nu = p + \frac{1}{2}$ ,  $k = k_\nu$ ,  $\sigma_f = 1$ . Fix a unit vector  $v$ ,  $\gamma \in \mathbb{R}$ ,  $d \geq 2$ , and  $\ell \geq \max\{\sqrt{2\nu}d, \zeta_1(p)/\sqrt{2\nu}\}$ . Set  $\zeta_0 := \sqrt{2\nu}\ell$ ,*

$$\beta := \frac{\gamma \ell^2 e^{\zeta_0}}{2\nu Q^{(2)}(\zeta_0)}, \quad \psi_{v,\gamma}(x) := \beta[k(x, \ell^2 v) + k(x, -\ell^2 v)].$$

*Then for every  $x \in [-1, 1]^d$ , with  $t = v^\top x$  and  $C_0 := 2\beta\rho_\nu(\zeta_0)$ ,*

$$|\psi_{v,\gamma}(x) - C_0 - \gamma t^2| \leq C(p) \frac{d^2}{\ell}$$

*for a constant  $C(p)$  depending only on  $p$ , not on  $d, \ell, \gamma$ .*

*Proof.* Define

$$\tau := \sqrt{2\nu}t/\ell, \quad \zeta_{\pm} := \sqrt{2\nu}r_{\pm}/\ell, \quad r_{\pm} := \|z \mp Rv\|.$$

Since  $\ell^2 \geq d^2 \geq 2d$ , we can invoke Lemma 3.2 with  $R = \ell^2$  and  $D = d$ , giving

$$\zeta_{\pm} = \zeta_0 \mp \tau + \eta''_{\pm}, \quad |\eta''_{\pm}| = \frac{\sqrt{2\nu}|\eta_{\pm}|}{\ell} \leq \frac{2\sqrt{2\nu}d^2}{\ell^3}.$$

By the choice of  $\ell$ , for  $d \geq 2$ ,

$$|\tau| = \frac{\sqrt{2\nu}d}{\ell} \leq 1, \quad |\eta''_{\pm}| \leq \frac{3}{8\nu d^2} < 1.$$

In addition,  $\zeta_0 \geq \zeta_1(p) \geq 2$ , so every argument  $\zeta_0 \mp \tau, \zeta_{\pm}$  lies in  $[\zeta_0 - 2, \zeta_0 + 2]$ , on which, by Lemma 3.5,

$$|\rho_{\nu}^{(j)}(\xi)| = |Q^{(j)}(\xi)|e^{-\xi} \leq A(p)(\zeta_0 + 3)^p e^2 e^{-\zeta_0}$$

for  $0 \leq j \leq 4$ . Let  $g(\tau) := \rho_{\nu}(\zeta_0 - \tau) + \rho_{\nu}(\zeta_0 + \tau)$ .  $g$  is even, so Taylor's theorem gives

$$g(\tau) = 2\rho_{\nu}(\zeta_0) + \rho''_{\nu}(\zeta_0)\tau^2 + \frac{1}{24}g^{(4)}(\xi)\tau^4$$

for some  $|\xi| \leq |\tau| \leq 1$  with

$$|g^{(4)}(\xi)| \leq 2A(p)(\zeta_0 + 3)^p e^2 e^{-\zeta_0}.$$

By the mean value theorem, replacing  $\zeta_{\pm}$  by  $\zeta_0 \mp \tau$  costs

$$|\rho_{\nu}(\zeta_{\pm}) - \rho_{\nu}(\zeta_0 \mp \tau)| \leq |\eta''_{\pm}|A(p)(\zeta_0 + 3)^p e^2 e^{-\zeta_0}.$$

Combining,

$$|\rho_{\nu}(\zeta_+) + \rho_{\nu}(\zeta_-) - 2\rho_{\nu}(\zeta_0) - \rho''_{\nu}(\zeta_0)\tau^2| \leq A(p)(\zeta_0 + 3)^p e^2 e^{-\zeta_0} \left[ \frac{\tau^4}{12} + |\eta''_+| + |\eta''_-| \right].$$

By definition,  $\beta\rho''_{\nu}(\zeta_0)\tau^2 = \gamma t^2$ , and  $C_0 = 2\beta\rho_{\nu}(\zeta_0)$  as claimed. Since  $Q^{(2)}(\zeta_0) \geq a(p)\zeta_0^p$  (Lemma 3.5),

$$\beta \leq \frac{|\gamma|\ell^2 e^{\zeta_0}}{2\nu a(p)\zeta_0^p},$$

so, using  $\zeta_0 \geq \zeta_1(p) \geq 2$  to guarantee  $(\zeta_0 + 3)/\zeta_0 \leq 4$ ,

$$\beta \cdot A(p)(\zeta_0 + 3)^p e^2 e^{-\zeta_0} \leq \frac{A(p)e^{24p}}{2\nu a(p)} |\gamma|\ell^2 =: K(p)|\gamma|\ell^2.$$

Finally,

$$\frac{\tau^4}{12} \leq \frac{d^4}{192\ell^4}, \quad |\eta''_{\pm}| \leq \frac{3\sqrt{2\nu}d^2}{\ell^3},$$

both  $O(d^2/\ell^3)$  using  $d^2 = O(\ell)$ , so

$$|\psi_{v,\gamma}(x) - C_0 - \gamma t^2| \leq K(p)|\gamma|\ell^2 \cdot O(d^2/\ell^3) = O_p(|\gamma|d^2/\ell),$$

our desired result. □

### 3.5 Polynomial-size approximation, unified

**Theorem 3.7** (Polynomial-size approximation). *Let  $k \in \{k_{\text{SE}}\} \cup \{k_\nu : \nu = \frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \dots\}$ , and let  $\kappa$  denote the corresponding implicit constant of Lemma 3.3 ( $\kappa = 5$ ) or Lemma 3.6 ( $\kappa = C(p)$ ). There is a training set  $X_2$  of size  $2m' = O(d^2)$ , computable in polynomial time from  $G$ , together with a length scale  $\ell := \max\{d, \sqrt{2\nu}d, \zeta_1(p)/\sqrt{2\nu}, 4\kappa\Gamma d^4\} = O(d^6)$  (the first three terms present, and vacuous, only in the Matérn case), such that the resulting posterior mean  $\mu_2$  satisfies*

$$|\mu_2(x) - C - \tilde{Q}(x)| \leq \frac{1}{4} \quad \text{for every } x \in [-1, 1]^d,$$

where  $C := \sum_k C_{0,k}$  is an explicit constant ( $C_{0,k} = 2\gamma_k\ell$  for SE, or the  $C_0$  of Lemma 3.6 per direction for Matérn).

*Proof.* Let  $\mu_2 := \sum_{k=1}^{m'} \psi_{v_k, \gamma_k}$ , using the  $m' = d + 2|E| = O(d^2)$  directions and coefficients of the decomposition of  $\tilde{Q}$  in §3.2, each realized by the two-point gadget of Lemma 3.3 (resp. 3.6).  $X_2$  consists of the  $2m'$  points  $x_0 \pm Rv_k$ , and  $\mu_2$  is the posterior mean for the dual coefficients  $\beta_k$  repeated on each pair. By the per-direction bound  $\kappa|\gamma_k|n^2/\ell$  and the triangle inequality, the total error is at most  $\kappa\Gamma d^2/\ell \cdot m' = O(\kappa\Gamma d^4/\ell)$  because  $|\gamma_k| \leq \Gamma \leq d^2$ , and  $\sum_k \gamma_k(v_k^\top z)^2 = \tilde{Q}(x)$  by construction. With  $\ell \geq 4\kappa\Gamma d^4$ , the right-hand side is at most  $\frac{1}{4}$ .  $\square$

### 3.6 Rational realizability, unified

The construction of §3.5 involves irrational quantities, and evaluating the kernel itself is transcendental even at rational points. All of this can be replaced by an explicit rational instance, computable in polynomial time, without materially changing the error bound. We isolate the one piece of algebra the squared exponential and Matérn kernels all share into a single lemma, so that only a few kernel-specific facts need checking per family.

**Lemma 3.8** (General rational realizability). *Let  $k$  be a stationary isotropic strictly PD kernel which is  $L$ -Lipschitz in each argument. Let  $X_2 = \{p_1, \dots, p_N\} \subset \mathbb{R}^d$  be our data points and  $\beta \in \mathbb{R}^N$  with  $\|\beta\|_2 \leq B$  be dual coefficients, with  $N$  even and points paired as in §3.5. Define  $\mu_2(x) := k(x, X_2)^\top \beta$ . Suppose that for every requested bit-precision  $b$  beyond an explicit threshold, one can compute the following in time  $\text{poly}(n, b)$ :*

- rational points  $\hat{p}_k \in \mathbb{Q}^d$  with  $\|\hat{p}_k - p_k\| \leq \delta_p$ ,
- rational coefficients  $\hat{\beta}_k \in \mathbb{Q}$  with  $|\hat{\beta}_k - \beta_k| \leq 2^{-b}$ , and
- a rational symmetric  $\bar{K} \in \mathbb{Q}^{N \times N}$  with  $|\bar{K}_{pq} - k(\hat{p}_p, \hat{p}_q)| \leq 2^{-b}$ .

Set  $\hat{X}_2 := \{\hat{p}_k\}$ ,  $\hat{K} := k(\hat{X}_2, \hat{X}_2)$ , and  $\hat{y}_2 := (\bar{K} + I)\hat{\beta}$ , repeating  $\hat{\beta}_k$  on its pair, and let

$$\hat{\mu}_2(x) := k(x, \hat{X}_2)^\top (\hat{K} + I)^{-1} \hat{y}_2$$

be the true posterior mean of this rational instance. Defining

$$\delta_2 := L\delta_p, \quad \delta_1 := N2^{-b}, \quad \delta_3 := \sqrt{N}2^{-b}, \quad A := (1 + \delta_1)(B + \delta_3),$$

we have

$$\sup_{x \in [-1, 1]^d} |\hat{\mu}_2(x) - \mu_2(x)| \leq \sqrt{N}[\delta_2 A + \delta_1(B + \delta_3) + \delta_3].$$

*Proof.* Since  $k$  is strictly PD,  $\hat{K} \succeq 0$ , so  $\|(\hat{K} + I)^{-1}\|_{\text{op}} \leq 1$ . It follows that

$$(\hat{K} + I)^{-1}(\bar{K} + I) = I + E \quad \text{with} \quad \|E\|_{\text{op}} \leq \|\bar{K} - \hat{K}\|_{\text{op}} \leq \|\bar{K} - \hat{K}\|_{\text{F}} \leq N2^{-b} = \delta_1.$$

Defining

$$\alpha_2 := (\hat{K} + I)^{-1}\hat{y}_2 = \hat{\beta} + E\hat{\beta},$$

we get  $\|\alpha_2 - \hat{\beta}\| \leq \delta_1\|\hat{\beta}\|$ . Since  $\|\hat{\beta} - \beta\| \leq \sqrt{N}2^{-b} = \delta_3$  and  $\|\beta\| \leq B$ , we have  $\|\hat{\beta}\| \leq B + \delta_3$ , so

$$\|\alpha_2\| \leq (1 + \delta_1)(B + \delta_3) = A \quad \text{and} \quad \|\alpha_2 - \beta\| \leq \|\alpha_2 - \hat{\beta}\| + \|\hat{\beta} - \beta\| \leq \delta_1(B + \delta_3) + \delta_3.$$

Splitting

$$\hat{\mu}_2(x) - \mu_2(x) = [k(x, \hat{X}_2) - k(x, X_2)]^\top \alpha_2 + k(x, X_2)^\top (\alpha_2 - \beta)$$

and applying Cauchy–Schwarz to each term, using the pointwise Lipschitz bound

$$|k(x, \hat{p}_k) - k(x, p_k)| \leq L\delta_p = \delta_2$$

on each of the  $N$  coordinates, and using  $|k(x, p_k)| \leq \sigma_f^2 = 1$  so  $\|k(x, X_2)\| \leq \sqrt{N}$ , we obtain our desired result.  $\square$

**Remark 3.9** (Choosing  $b$ ). *In every instance below,  $\delta_p = D_0 2^{-b}$  for an explicit construction-dependent  $D_0$ , and  $B \leq P_1(d)e^{c\ell}$  for an explicit polynomial  $P_1$  and kernel-dependent constant  $c$ . Then the bound of Lemma 3.8 is at most  $P(d)e^{c\ell}2^{-b}$  for an explicit polynomial  $P$  collecting  $N, L, D_0, P_1$ ; since  $e^{c\ell} < 2^{\lceil c\ell/\ln 2 \rceil}$ , choosing  $b := \lceil c\ell/\ln 2 \rceil + \lceil \log_2(8P(d)) \rceil$  gives the bound  $\leq \frac{1}{8}$ , with  $b = O(\ell) + O(\log d) = O(\ell)$ , polynomial in  $d$  since  $\ell = O(d^6)$  (Theorem 3.7).*

**Corollary 3.10** (Rational realizability, SE). *The construction of Lemma 3.3/Theorem 3.7 for  $k_{\text{SE}}$  admits an explicit rational instance, computable in time polynomial in  $d$ , whose true posterior mean differs from  $\mu_2$  by at most  $\frac{1}{8}$  uniformly on  $[-1, 1]^d$ .*

*Proof.* Fix  $\sigma_n^2 := 1$  (harmless by fact (a) of §2.1), and since  $K \succeq 0$ ,  $\|(K + I)^{-1}\|_{\text{op}} \leq 1$ . The squared-exponential kernel is known to have Lipschitz constant  $\frac{1}{\ell}e^{-1/2} < \frac{1}{\ell}$ , so we can take  $L := 1/\ell$ . Using Brent’s algorithm ([1]) for  $\sqrt{\cdot}$  and  $\exp(\cdot)$  to  $b$  bits in time polynomial in  $b$ , construct the following:

- $\hat{v}_{ij}^\pm := \hat{s}(e_i \pm e_j)$ ,
- $\hat{s} \in \mathbb{Q}$  with  $|\hat{s} - 1/\sqrt{2}| \leq 2^{-b}$ ,
- $\hat{R} \in \mathbb{Q}$  with  $|\hat{R} - \ell^{3/2}| \leq 2^{-b}$ ,
- $\hat{\beta}_k \in \mathbb{Q}$  with  $|\hat{\beta}_k - \beta_k| \leq 2^{-b}$  for  $\beta_k = \gamma_k \ell e^{\ell/2}$ , and
- $\bar{K} \in \mathbb{Q}^{N \times N}$  with  $|\bar{K}_{pq} - k_{\text{SE}}(\hat{p}_p, \hat{p}_q)| \leq 2^{-b}$  for  $\hat{p}_k := x_0 \pm \hat{R}\hat{v}_k$ .

Then

$$\delta_p \leq \hat{R}\|\hat{v}_k - v_k\| + |\hat{R} - R| \leq 4\ell^{3/2}2^{-b} =: D_0 2^{-b}$$

since  $\hat{R} \leq \ell^{3/2} + 1 \leq 2\ell^{3/2}$ , and  $\|\beta\|_\infty \leq \Gamma \ell e^{\ell/2}$ , so  $B \leq \sqrt{N}\Gamma \ell e^{\ell/2} = P_1(d)e^{\ell/2}$  with  $c = 1/2$ . Applying Remark 3.9 with  $L = 1/\ell$ ,  $D_0 = 4\ell^{3/2}$ , we see that  $b = O(\ell) = O(d^6)$  gives error  $\leq \frac{1}{8}$ , and every quantity involved has  $O(\ell + \log d) = O(\ell)$  bits, so the whole instance is polynomial-time constructible.  $\square$

**Corollary 3.11** (Rational realizability, every half-integer Matérn kernel). *The construction of Lemma 3.6/Theorem 3.7 for  $k_\nu$  admits an explicit rational instance, computable in time polynomial in  $d$ , whose true posterior mean differs from  $\mu_2$  by at most  $\frac{1}{8}$  uniformly on  $[-1, 1]^d$ .*

*Proof.* Fix  $\sigma_n^2 := 1$  as above ( $K \succeq 0$  holds for  $k_\nu$  too, by strict PD). Writing

$$k_\nu(x, p) = \rho_\nu(\zeta), \quad \zeta = \frac{\sqrt{2\nu}\|x - p\|}{\ell}, \quad \nabla_p k_\nu = \frac{\rho'_\nu(\zeta)\sqrt{2\nu}}{\ell} \cdot \frac{x - p}{\|x - p\|},$$

and

$$|\rho'_\nu(\zeta)| = |Q_1(\zeta)|e^{-\zeta} \leq M_1(p) := \sup_{\zeta \geq 0} |Q^{(1)}(\zeta)|e^{-\zeta} < \infty \Rightarrow L := \frac{\sqrt{2\nu}M_1(p)}{\ell}.$$

The directions  $v_k$  round exactly as in Corollary 3.10. The offset  $R = \ell^2$  is already an exact integer, so no offset-rounding is needed. For

$$\beta_k = \frac{\gamma_k \ell^2 e^{\zeta_0}}{2\nu Q^{(2)}(\zeta_0)},$$

do the following:

- construct  $\hat{\zeta}_0 \in \mathbb{Q}$  with  $|\hat{\zeta}_0 - \zeta_0| \leq 2^{-b'}$  via one Brent-sqrt call ( $\zeta_0 = \sqrt{2\nu}\ell$ ),
- evaluate  $Q^{(2)}(\hat{\zeta}_0)$  exactly by rational arithmetic to precision matching  $b'$ , and
- evaluate  $e^{\hat{\zeta}_0}$  via one Brent-exp call to precision matching  $b'$ .

Since  $Q^{(2)}(\zeta_0) \geq a(p)\zeta_0^p$  is bounded away from zero and  $Q^{(2)'} has degree  $p$ , both by Lemma 3.5, the composed error propagates to$

$$|\hat{\beta}_k - \beta_k| \leq 2^{-b}, \quad b' = b + O(\log(\Gamma \ell^2 e^{\zeta_0}/a(p))) = b + O(\zeta_0 + \log d),$$

which is polynomial in  $b + \ell$ . The kernel matrix entries  $k_\nu(\hat{p}_p, \hat{p}_q)$  are one Brent-sqrt (distance) composed with the exact polynomial  $Q_p$  and one Brent-exp, propagated to  $2^{-b}$  overall by the same composition-of-Brent-calls argument, using the  $O(1/\ell)$ -type Lipschitz bound above on the distance argument. So

$$\|\beta\|_\infty \leq \frac{\Gamma \ell^2 e^{\zeta_0}}{2\nu a(p)\zeta_0^p} =: P_1(d)e^{\sqrt{2\nu}\ell}.$$

Applying Remark 3.9,  $b = O(\ell) = O(d^6)$  gives error  $\leq \frac{1}{8}$ , polynomial-time constructible.  $\square$

### 3.7 Main theorem

**Theorem 3.12** (NP-hardness over the box). *Let  $k$  be  $k_{SE}$  or  $k_\nu$  for any half-integer  $\nu = p + \frac{1}{2}$ . There is a polynomial-time reduction mapping any unweighted MAX-CUT instance  $G$  (or, by Remark 3.13, any weighted instance with  $W_{\text{tot}} = \text{poly}(d)$ ) to a Gaussian process (training set  $X_2$ , targets  $y_2$ , hyperparameter  $\ell = O(d^6)$ ) such that, letting  $\mu_2$  be the resulting posterior mean,*

$$Q^* = \text{round}\left(\min_{x \in [-1, 1]^d} \mu_2(x) - C + \Lambda d\right),$$

where  $C, \Lambda$  are explicit, polynomial-time computable constants. Consequently, minimizing  $\mu_2(x)$  to any fixed accuracy better than  $\frac{1}{2}$  is NP-hard, and so is minimizing the posterior mean of any strictly PD, stationary, isotropic kernel over  $[-1, 1]^d$  that admits such a construction; e.g., SE and every half-integer Matérn kernel.

*Proof.* By Theorem 3.7 and Corollary 3.10 (resp. Corollary 3.11),  $|\mu_2(x) - C - \tilde{Q}(x)| \leq \frac{1}{4} + \frac{1}{8} = \frac{3}{8}$  for all  $x \in [-1, 1]^d$ . A uniform bound on a perturbation bounds the perturbation of the minimum by the same amount, so

$$|\min_x \mu_2(x) - C - \min_x \tilde{Q}(x)| \leq \frac{3}{8}.$$

By Lemma 3.1,  $\min_x \tilde{Q}(x) = Q^* - \Lambda d$ . Since  $w$  is integer-valued,  $Q^* \in \mathbb{Z}$ , and  $C, \Lambda, d$  are known exactly,

$$|\min_x \mu_2(x) - C + \Lambda d - Q^*| \leq \frac{3}{8} < \frac{1}{2},$$

and rounding recovers  $Q^*$  exactly. Since computing  $Q^*$  is NP-hard, so is computing  $\min_x \mu_2(x)$  to within any accuracy strictly better than  $\frac{1}{2}$ .  $\square$

**Remark 3.13** (Weighted MAX-CUT). *The unweighted standing assumption is not essential to the argument, only to the claim that  $\ell$  is polynomial in the bit-length of the input. For weights given in unary, or promised to satisfy  $W_{\text{tot}} \leq \text{poly}(d)$ , the identical construction with  $\Lambda = \frac{1}{2}W_{\text{tot}}$ ,  $\Gamma = \max(4\Lambda, 2\max_{ij} w_{ij}) \leq 2W_{\text{tot}}$ , and  $\ell = \max(d, 4\kappa\Gamma d^4) = \text{poly}(d)$  goes through verbatim, and Theorem 3.12 holds for such instances too. For general binary-encoded weights,  $W_{\text{tot}}$  (hence  $\ell$ ) can be exponential in the input size, and this reduction no longer runs in polynomial time; whether a different construction removes this restriction is open.*

## 4 Scope: how general is this?

Theorem 3.12 covers SE and every half-integer Matérn kernel. It is worth being precise about which structural properties of a stationary isotropic kernel this argument actually uses, so as to know how much further it reaches.

Tracing through §3.4–§3.6, a stationary isotropic kernel  $k_\ell(x, x') = \rho(\|x - x'\|/\ell)$  (Definition 2.1) supports the same construction provided the shape function  $\rho$  admits, for arbitrarily large  $\zeta_0$ , either:

- an *exact* algebraic factorization of  $\rho(r_+) + \rho(r_-)$  into a product of an isotropic decay factor and an even function of  $t = v^\top z$  — the SE mechanism (Lemma 3.3), available whenever  $\rho$  is itself a function of  $\|\cdot\|^2$  rather than  $\|\cdot\|$ , since then  $r_\pm^2 = \|z\|^2 + R^2 \mp 2Rt$  needs no Taylor approximation at all; or
- a *local self-similarity of derivatives*: constants  $\zeta_1, A$  (depending only on the shape of  $\rho$ ) such that  $\rho''(\zeta_0) \neq 0$  for all  $\zeta_0 \geq \zeta_1$ , and  $|\rho^{(j)}(\zeta)| \leq A|\rho''(\zeta_0)|$  for  $0 \leq j \leq 4$  and  $|\zeta - \zeta_0| \leq 2$  — the Matérn mechanism (Lemma 3.6), which needs Lemma 3.2’s approximate distance expansion because  $\rho$  is a function of  $\|\cdot\|$  itself.

Together with  $\rho \in C^4((0, \infty))$  and genuine decay  $\rho(\zeta) \rightarrow 0$ , this is what makes  $\beta$  (chosen to cancel  $\rho''(\zeta_0)$  exactly) well defined, makes the remaining derivatives (needed for the Taylor remainder) comparably sized, and makes  $C_0 = 2\beta\rho(\zeta_0)$  an explicit, finite, exactly computable constant.

*What else satisfies this, without full proof here.* Non-half-integer Matérn ( $\nu \notin \mathbb{Z} + \frac{1}{2}$ ) replaces  $Q_p(\zeta)$  with the large-argument asymptotic series of the Bessel- $K$  function,  $K_\nu(\zeta) \sim \sqrt{\pi/2\zeta} e^{-\zeta}(1 + O(1/\zeta))$ , which has the same qualitative shape (polynomial-type prefactor times  $e^{-\zeta}$ ) and plausibly satisfies the self-similarity envelope with an asymptotic-series argument replacing Lemmas 3.4–3.5; we do not carry this out. The rational quadratic kernel,  $\rho(r) = (1 + r^2/2\alpha\ell^2)^{-\alpha}$ , is (like SE) a function of squared distance, so the exact-factorization route of Lemma 3.3 does not directly apply (there is no  $e^{a+b} = e^a e^b$  identity for a power), but it decays polynomially rather than exponentially

and never has vanishing curvature, so *some* variant of the argument plausibly applies with different error bookkeeping; we do not spell this out either.

*What provably fails.* Compactly supported kernels (e.g., Wendland functions), where  $\rho(\zeta) = 0$  identically for  $\zeta$  beyond a fixed support radius, break the construction at its source: for any  $\zeta_0$  past the support radius,  $\rho''(\zeta_0) = 0$  and  $\rho(\zeta_0) = 0$ , so  $\beta$  (which divides by  $\rho''(\zeta_0)$ ) is not merely hard to bound but undefined, and the two-point gadget realizes the zero function. This is not a gap in the proof technique that a cleverer choice of  $\zeta_0$  could close because every  $\zeta_0$  past the support radius is equally degenerate, so a fundamentally different construction, not merely a different envelope lemma, would be needed for such kernels, if hardness holds for them at all.

Kernels that do not decay to 0, such as periodic kernels, leave a genuine open question. Non-stationary kernels do not fit the framework of Definition 2.1 at all, so the question of hardness for them is a different one, not addressed by any extension of this argument.

## 5 Extension to acquisition functions

We now show the hardness of Theorem 3.12 is not special to the bare posterior mean: it persists for standard BO acquisition functions built from  $(\mu, \sigma)$ , because  $\sigma$  is always confined to a fixed compact range that is completely decoupled from the combinatorial signal carried by  $\mu$ .

### 5.1 A domination lemma

**Lemma 5.1** (Scale domination). *Let  $\mu_2$  be as in Theorem 3.7, built from training set  $X_2$  and targets  $y_2$ . For  $c_{\text{scale}} > 0$ , let  $\mu_2^{(c_{\text{scale}})} := c_{\text{scale}}\mu_2$  be the posterior mean from targets  $c_{\text{scale}}y_2$  on the same  $X_2$ , and let  $\sigma$  be the posterior standard deviation for  $X_2$  (independent of  $c_{\text{scale}}$ , by fact (b) of §2.1). Let  $h : [0, \sigma_f] \rightarrow \mathbb{R}$  satisfy  $|h(\sigma)| \leq H$  for all  $\sigma$ , with  $H$  independent of  $c_{\text{scale}}$ ,  $G$ ,  $w$ . If  $c_{\text{scale}} \geq 8(H + 1)$ , then  $\arg \min_x [\mu_2^{(c_{\text{scale}})}(x) - h(\sigma(x))]$  recovers  $Q^*$  via the identical rounding argument of Theorem 3.12 (applied to the rescaled construction).*

*Proof.* By Theorem 3.7 and Corollaries 3.10–3.11 applied with targets scaled by  $c_{\text{scale}}$  (both bounds scale linearly),

$$|\mu_2^{(c_{\text{scale}})}(x) - c_{\text{scale}}C - c_{\text{scale}}\tilde{Q}(x)| \leq \frac{3c_{\text{scale}}}{8}$$

for all  $x$ . Hence

$$|\mu_2^{(c_{\text{scale}})}(x) - h(\sigma(x)) - c_{\text{scale}}C - c_{\text{scale}}\tilde{Q}(x)| \leq \frac{3c_{\text{scale}}}{8} + H \leq \frac{c_{\text{scale}}}{2} - 1 < \frac{c_{\text{scale}}}{2}$$

using  $H \leq c_{\text{scale}}/8 - 1$  (the strict form of the hypothesis, since  $\frac{3}{8} + \frac{1}{8} = \frac{1}{2}$  leaves no slack on its own). Dividing by  $c_{\text{scale}}$ , the function  $x \mapsto \frac{1}{c_{\text{scale}}}[\mu_2^{(c_{\text{scale}})}(x) - h(\sigma(x))]$  is within  $\frac{1}{2}$  of  $C + \tilde{Q}(x)$  uniformly, so Theorem 3.12's rounding argument applies.  $\square$

The content of Lemma 5.1 is that any acquisition function of the form  $\mu(x) - h(\sigma(x))$ , with  $h$  bounded uniformly over  $[0, \sigma_f]$  by an instance-independent constant  $H$ , inherits NP-hardness for free.

### 5.2 Lower confidence bound

**Corollary 5.2** (LCB). *For any fixed  $\kappa \geq 0$ , minimizing  $\text{LCB}_\kappa(x) := \mu(x) - \kappa\sigma(x)$  over  $[-1, 1]^d$  is NP-hard.*

*Proof.* Apply Lemma 5.1 with  $h(\sigma) = \kappa\sigma$ ,  $H = \kappa\sigma_f$ .  $\square$

### 5.3 Expected improvement

Let  $\tilde{\Phi}(z) := z\Phi(z) + \phi(z)$  for the standard normal CDF/PDF  $\Phi, \phi$ ; then  $\tilde{\Phi}'(z) = \Phi(z)$ , so  $\tilde{\Phi}$  is strictly increasing, with  $\tilde{\Phi}(z) \rightarrow 0^+$  as  $z \rightarrow -\infty$  and  $\tilde{\Phi}(z) - z \rightarrow 0$  as  $z \rightarrow +\infty$ . Expected improvement, for incumbent  $f_{\text{best}}$ , is  $\text{EI}(x) = \sigma(x)\tilde{\Phi}(\gamma(x))$ ,  $\gamma(x) = (\mu(x) - f_{\text{best}})/\sigma(x)$ , extended continuously to  $\sigma(x) = 0$  by  $\text{EI}(x) := \max(\mu(x) - f_{\text{best}}, 0)$ . EI rewards large  $\mu$ , so we apply it to  $\nu_2 := -\mu_2$ , itself a posterior mean on  $X_2$  from targets  $-y_2$ .

**Lemma 5.3** (Two-sided EI tail bound). *For every  $z > 0$ ,  $0 \leq \tilde{\Phi}(z) - z \leq \phi(z)$ .*

*Proof.*  $\tilde{\Phi}(z) - z = \phi(z) - z\Phi(-z)$ . The Mills-ratio bound  $\Phi(-z) \leq \phi(z)/z$  for  $z > 0$  gives  $z\Phi(-z) \leq \phi(z)$ , so  $\tilde{\Phi}(z) - z \geq 0$ ; and  $z\Phi(-z) \geq 0$  gives  $\tilde{\Phi}(z) - z \leq \phi(z)$ .  $\square$

**Theorem 5.4** (Expected improvement). *Let  $\mu_2, X_2, y_2, \ell$  be as in Theorem 3.12 (so  $|\mu_2(x) - C - \tilde{Q}(x)| \leq \frac{3}{8}$  for all  $x$ ),  $\nu_2 := -\mu_2$ ,  $M := W_{\text{tot}} + \Lambda d$ ,  $L_0 := -C - M - \frac{3}{8}$ . For any  $c_{\text{scale}} \geq 2$ , apply EI to  $c_{\text{scale}}\nu_2$  with incumbent  $f_{\text{best}} := c_{\text{scale}}(L_0 - 1)$ . Then every  $x^* \in \arg \max_{x \in [-1, 1]^d} \text{EI}(x)$  satisfies  $Q^* = \text{round}(\mu_2(x^*) - C + \Lambda d)$ : maximizing EI and rounding recovers  $Q^*$ , so maximizing EI over  $[-1, 1]^d$  is NP-hard.*

*Proof.* Since  $u(x) \in [-1, 1]^n$ ,

$$|\tilde{Q}(x)| \leq W_{\text{tot}} + \Lambda n = M,$$

so

$$\mu_2(x) \leq C + M + \frac{3}{8}, \quad \nu_2(x) \geq L_0 \quad \forall x.$$

Writing

$$\Delta(x) := c_{\text{scale}}\nu_2(x) - f_{\text{best}} = c_{\text{scale}}(\nu_2(x) - L_0 + 1),$$

$\Delta(x) \geq c_{\text{scale}} > 0$  for every  $x$ . For  $\sigma(x) > 0$ ,

$$\gamma(x) = \Delta(x)/\sigma(x) \geq \Delta(x) \geq c_{\text{scale}}$$

as  $\sigma \leq \sigma_f = 1$ , so by Lemma 5.3 and monotonicity of  $\phi$  on  $[0, \infty)$ ,

$$0 \leq \text{EI}(x) - \Delta(x) \leq \sigma(x)\phi(\gamma(x)) \leq \phi(c_{\text{scale}});$$

at  $\sigma(x) = 0$  this holds too by the continuous extension. So  $|\text{EI}(x) - \Delta(x)| \leq \phi(c_{\text{scale}}) =: \epsilon(c_{\text{scale}})$  uniformly. Let

$$x^* \in \arg \max_x \text{EI}(x), \quad x^\sharp \in \arg \max_x \nu_2(x);$$

both exist by continuity/compactness. Since  $\Delta$  is affine increasing in  $\nu_2$  with slope  $c_{\text{scale}}$  and  $\text{EI}(x^*) \geq \text{EI}(x^\sharp)$ ,

$$c_{\text{scale}}(\nu_2(x^*) - \nu_2(x^\sharp)) \geq -2\epsilon(c_{\text{scale}}),$$

so

$$\mu_2(x^*) \leq \min_x \mu_2(x) + 2\epsilon(c_{\text{scale}})/c_{\text{scale}}.$$

The map  $c \mapsto 2\epsilon(c)/c$  is strictly decreasing on  $(0, \infty)$  and equals  $\phi(2) \approx 0.054 < \frac{1}{16}$  at  $c = 2$ , so  $2\epsilon(c_{\text{scale}})/c_{\text{scale}} \leq \frac{1}{16}$  for every  $c_{\text{scale}} \geq 2$ . Combined with

$$|\min_x \mu_2(x) - (C + Q^* - \Lambda n)| \leq \frac{3}{8}$$

(Theorem 3.12's proof),

$$|\mu_2(x^*) - (C + Q^* - \Lambda n)| \leq \frac{3}{8} + \frac{1}{16} = \frac{7}{16} < \frac{1}{2},$$

and rounding recovers  $Q^*$ .  $\square$

**Remark 5.5.** *Theorem 5.4 needs no limit as  $c_{\text{scale}} \rightarrow \infty$ : Lemma 5.3 is exact and two-sided, so a single explicit threshold  $c_{\text{scale}} \geq 2$  suffices, with every instance-dependent quantity folded into  $f_{\text{best}}$ .*

## 5.4 Gittins index

**Corollary 5.6** (Gittins index). *Fix  $c_0 > 0$ . Minimizing  $g(x) := \mu(x) - \sigma(x)\tilde{\Phi}^{-1}(c_0/\sigma(x))$  over  $[-1, 1]^d$  is NP-hard.*

*Proof.*  $\tilde{\Phi} : \mathbb{R} \rightarrow (0, \infty)$  is a strictly increasing bijection, so  $\tilde{\Phi}^{-1} : (0, \infty) \rightarrow \mathbb{R}$  is well defined. Let  $h(\sigma) := \sigma\tilde{\Phi}^{-1}(c_0/\sigma)$  for  $\sigma \in (0, \sigma_f]$ . As  $\sigma \rightarrow 0^+$ ,  $c_0/\sigma \rightarrow \infty$ , and since  $\tilde{\Phi}(z) - z \rightarrow 0$  as  $z \rightarrow \infty$ ,  $\tilde{\Phi}^{-1}(w) - w \rightarrow 0$  as  $w \rightarrow \infty$ ; hence  $h(\sigma) = \sigma(c_0/\sigma + o(1)) \rightarrow c_0$ , so  $h$  extends continuously to  $[0, \sigma_f]$  with  $h(0) := c_0$ . A continuous function on a compact interval is bounded, with bound  $H$  depending only on  $c_0, \sigma_f$ . Apply Lemma 5.1.  $\square$

**Remark 5.7.** *Corollary 5.6 needs no bespoke tail-bound lemma of its own, since  $g(x) = \mu(x) - h(\sigma(x))$  is exactly of Lemma 5.1's form with no saturation as  $\mu \rightarrow \pm\infty$ . By contrast, probability of improvement,  $\text{PI}(x) = \Phi(\gamma(x)) \in [0, 1]$ , is bounded in  $x$  for every fixed scale, so sending  $c_{\text{scale}} \rightarrow \infty$  saturates every improving vertex to  $\text{PI} \approx 1$  and erases the distinctions needed to recover  $Q^*$ ; a finite, carefully tuned  $c_{\text{scale}}$  would be needed instead, and we do not carry out that argument here.*

## 6 Concluding remarks

The results above show that NP-hardness of minimizing  $\mu$ , LCB, EI, and maximizing GI over  $[-1, 1]^d$  is a structural consequence of pairing (i) a stationary, isotropic, strictly positive definite kernel whose far field admits either an exact quadratic-exponent factorization or a self-similar derivative envelope (§4) — equivalently, an infinite-dimensional RKHS rich enough to approximate an indefinite quadratic form to fixed accuracy with a polynomial-size kernel expansion — with (ii) an acquisition rule built from  $(\mu, \sigma)$  that does not saturate away the signal carried by  $\mu$ , since  $\sigma$  is always confined to  $[0, \sigma_f]$  independently of the training targets. This is the Gaussian-process analogue of the classical fact that concave (equivalently, indefinite) quadratic minimization over a polytope is NP-hard ([6]), proved by the same shift-to-concavity plus extreme-point argument used in Lemma 3.1; the work of §3.4–§3.6 is entirely in realizing that hard quadratic core as an honest finite kernel expansion, for an entire family of kernels at once rather than one at a time.

## References

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